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Requirements Definition

Nonfunctional Requirements

1. Operational Requirements
   - 1.1. The system will automatically back up every night.
   - 1.2. The system should integrate with other IT systems within the hospital, including the pharmacy inventory system and other aspects of the EHR system.
   - 1.3. The system will integrate with a Clinical Decision Support (CDS) system, which includes a database for medication/allergy interaction information.
   - 1.4. The system can operate on Google Chrome and Microsoft Edge web browsers.
2. Performance Requirements
   - 2.1. The system will process medication checks in 3 seconds or less.
   - 2.2. The system will retrieve medication orders within 2 seconds.
   - 2.3. Response time for interactions between the user and the system should be 1 second or less.
   - 2.4. Stat orders must be placed at the top of the processing queue within 2 seconds.
   - 2.5. The system must handle simultaneous medication orders from multiple physicians.
3. Security Requirements
   - 3.1. Only doctors can order medications.
   - 3.2. Only pharmacists can re-validate medication orders and place orders in the processing queue.
   - 3.3. Only pharmacists and pharmacy techs can process medication orders.
   - 3.4. Only nurses can document administration of a medication.
4. Cultural and Political Requirements
   - 4.1. The system must comply with HIPAA.

Functional Requirements

1. Order Medication
   - 1.1. Physician selects medication.
   - 1.2. Physician enters dosage.
   - 1.3. Physician submits the order.
2. Verify Medication Order
   - 2.1. The system checks that medications are within the appropriate dose range based on the patient's age and weight.
     - 2.1.1. If an incorrect dose is entered, the system will notify the physician and require the physician to re-enter an appropriate dose.
     - 2.1.2. If a correct dose is entered, the system will proceed with the order.

2.2. The system checks the medication for potential adverse reactions with the patient's known allergies.
    2.2.1. If an allergy conflict is detected, the system will notify the physician, and the physician can select an alternative medication or continue with the original medication.
    2.2.2. If no allergy conflict is detected, the system will proceed with the order.
2.3. The system checks the medication for potential adverse reactions with the patient's existing medications.
    2.3.1. If an adverse reaction is detected, the system will notify the physician, and the physician can select an alternative medication or continue with the original medication.
    2.3.2. If no adverse interaction is detected, the system will proceed with the order.
2.4. The system checks if the medication is in stock.
    2.4.1. If the medication is out of stock, the system will place an order for replenishment and notify the physician.
    2.4.2. If the medication if out of stock, the physician can cancel the order.
    2.4.3. If the medication is out of stock, the system will prompt the physician to select an alternative medication.
    2.4.4. If the medication is in stock, the system will proceed with the order.
2.5. The system will send validated medication orders to the pharmacy for processing.

3. Process Medication Order
    3.1. After re-checking the medication for compliance with dose, allergy, and medication interactions, the pharmacist must deem the order invalid or valid in the system.
        3.1.1. If the order is invalid, the system will return the order to the physician's worklist.
        3.1.2. If the order is valid, the pharmacist will place the order in the pharmacy tech's queue for processing.
    3.2. Pharmacy tech prepares the medication order and updates the order status in the system.
    3.3. Pharmacist checks the tech's order for accuracy and deems the order correct or incorrect in the system.
        3.3.1. If the order is incorrect, the system will send the order back to the tech for reprocessing.
4. Administer Medication
    4.1. Nurse can document administration of the medication to the patient.
        4.1.1. Nurse can document a witness.

4.1.2. Nurse must document amount of drug waste.
4.1.3. Nurse must document patient reaction to medication.
4.2. Nurse can document patient refusal of the medication.

Changes to Requirements Definition:

1. Separated “Order medication” use case into “Order medication” and “Verify medication order” use cases.
2. Added a separate CDS system as an actor within the CPOE system.
3. Added that the CPOE system will integrate with a CDS system in operational requirements.
4. Changed “The system will require the physician to select an alternative medication” to “The physician can select an alternative medication or continue with the original medication”.
5. Added “If the medication if out of stock, the physician can cancel the order”.
6. Added “If the medication is out of stock, the physician must indicate reason for cancellation”.
7. Moved “The system will send validated medication orders to the pharmacy for processing” from “Process Medication Order” use case to “Verify Medication Order” use case.
8. Added that the nurse can document patient refusal of medication.

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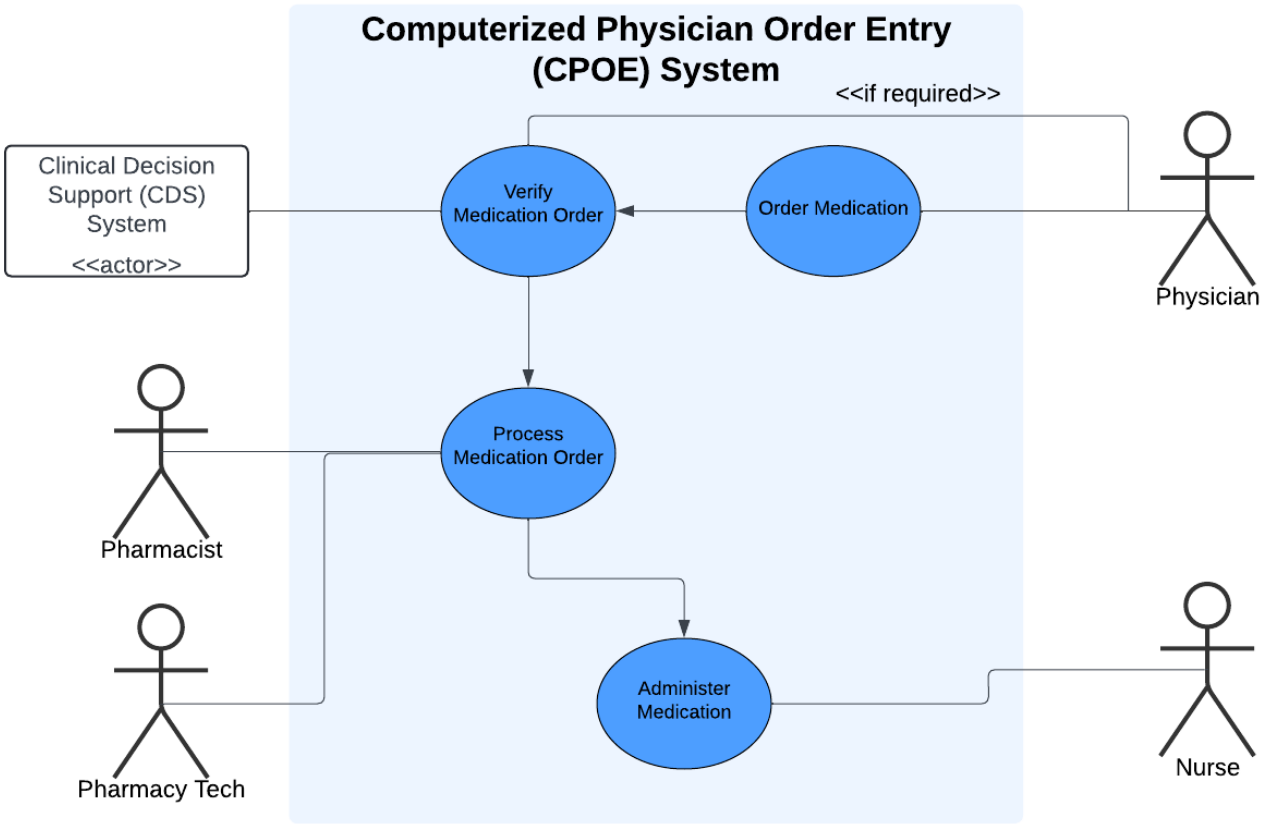

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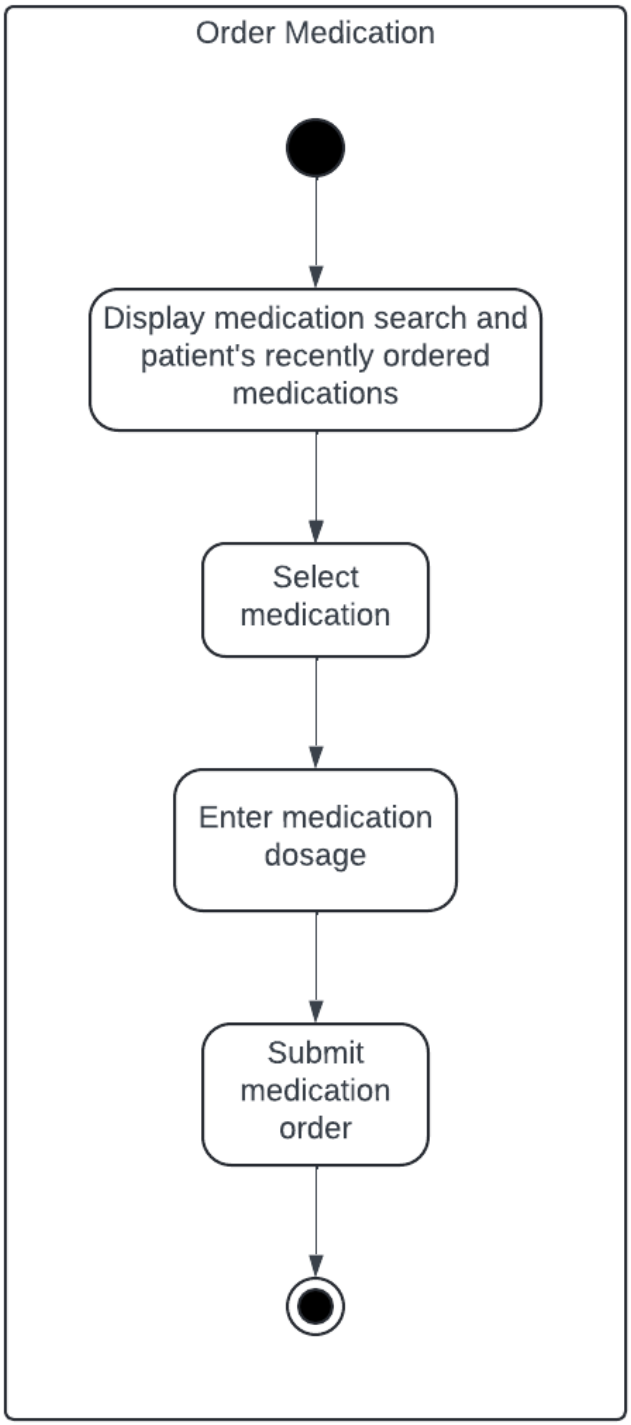

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Use Case Name: Order Medication	ID: 1	Importance Level: High
Primary Actor: Physician	Use Case Type: Detail, Essential	
Stakeholders and Interests: Physician – wants to easily order medications for patients CMO – wants to improve workflow and point-of-care medication ordering		
Brief Description: This use case describes how physicians order a medication for a patient.		
Triggers: Physician decides a patient needs a new medication or a patient needs a medication re-fill. Medication order was sent back to physician's worklist, so physician initiates new order. Type: External		
Relationships: Association: Physician Include: Extend: Generalization:		
Normal Flow of Events: 1. The system displays the medication search box and the patient's recently ordered medications. 2. The physician selects the medication to order. 3. The physician enters the medication dosage. 4. The physician submits the medication order.		
SubFlows:		
Alternate/Exceptional Flows:		

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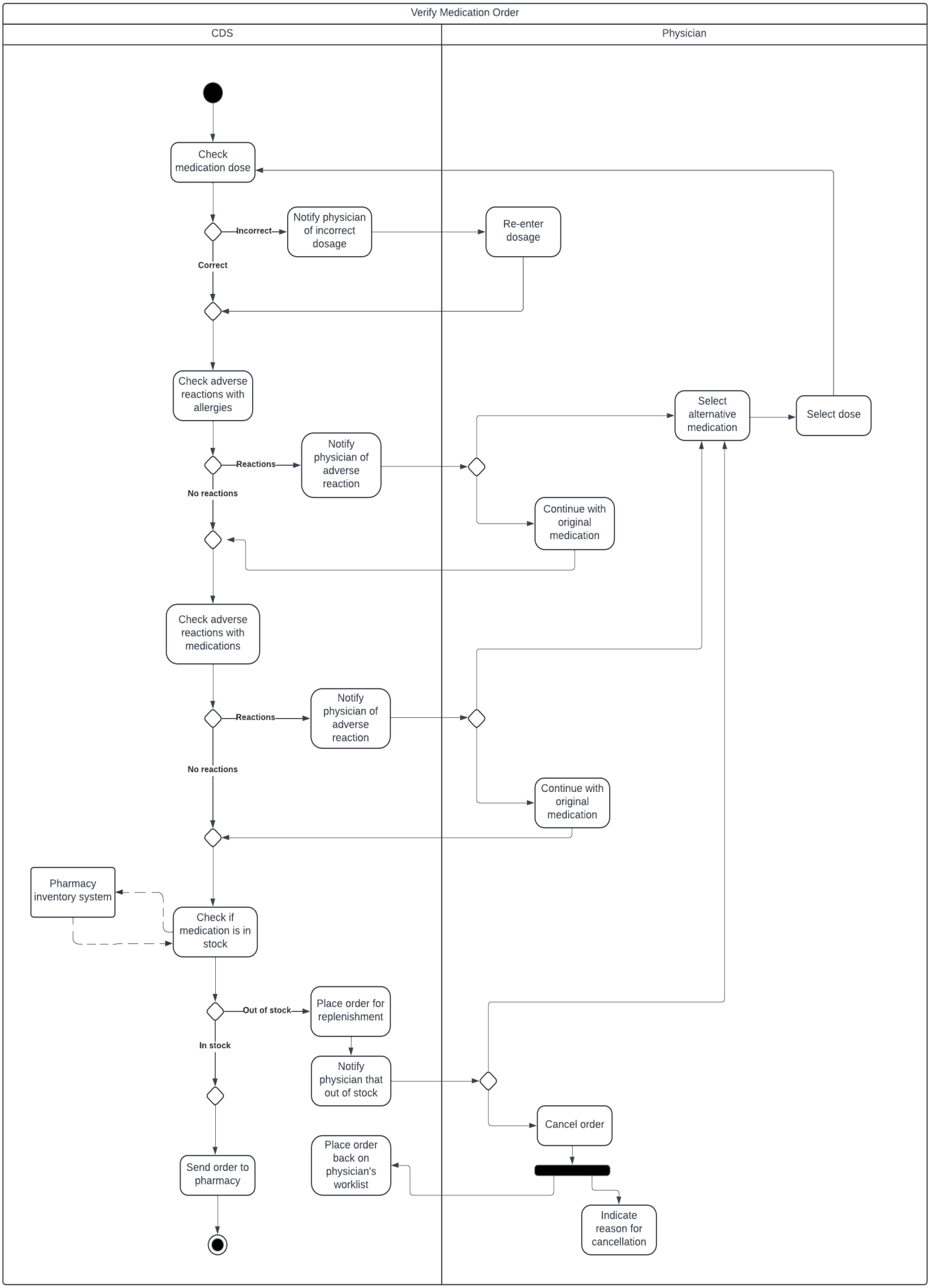

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Use Case Name: Verify Medication Order	ID: 2	Importance Level: Medium
Primary Actor: Clinical Decision Support (CDS) System	Use Case Type: Detail, Essential	

Stakeholders and Interests: Physician – wants to decrease mistakes in medication orders to ensure patient safety
Pharmacist – wants additional support in checking medication orders before arrival to the pharmacy

Brief Description: This use case describes how a medication order is checked for accuracy and safety by the CDS system and corrected by the physician if needed.

Trigger: The physician orders a medication.

Type: External

Relationships:
Association: CDS System and Physician
Include:
Extend:
Generalization:

Normal Flow of Events:

1. CDS checks the medication dose.
   If the medication dose is incorrect, S-1 subflow is performed.
2. CDS checks adverse reactions with allergies.
   If an adverse reaction is detected, the system notifies the physician of an adverse reaction.
   If the notified physician wants to select an alternative medication, S-2 subflow is performed.
   If the notified physician wants to continue with the original medication, the system proceeds.
3. CDS checks adverse reactions with medications.
   If an adverse reaction is detected, the system notifies the physician of an adverse reaction.
   If the notified physician wants to select an alternative medication, S-2 subflow is performed.
   If the notified physician wants to continue with the original medication, the system proceeds.
4. The system checks if the medication is in stock.
   If the medication is out of stock, the system places an order for replenishment and notifies the physician that the medication is out of stock.
   If the notified physician wants to select an alternative medication, S-2 subflow is performed.
   If the notified physician wants to cancel the order, S-3 subflow is performed, and the system does not proceed to step #5.
5. The system sends the order to the pharmacy.

SubFlows:

S-1: Re-enter dosage
1. The system notifies physician of incorrect dosage.
2. The physician re-enters the dose amount.

S-2: Select alternative medication
1. The physician selects an alternative medication
2. The physician selects a medication dose.
3. The system returns to step #1 in the normal flow.

S-3: Cancel order
1. The physician cancels the medication order.
2. The order is placed back on the physician's worklist.
3. The physician indicates the reason for cancellation.

Alternate/Exceptional Flows:

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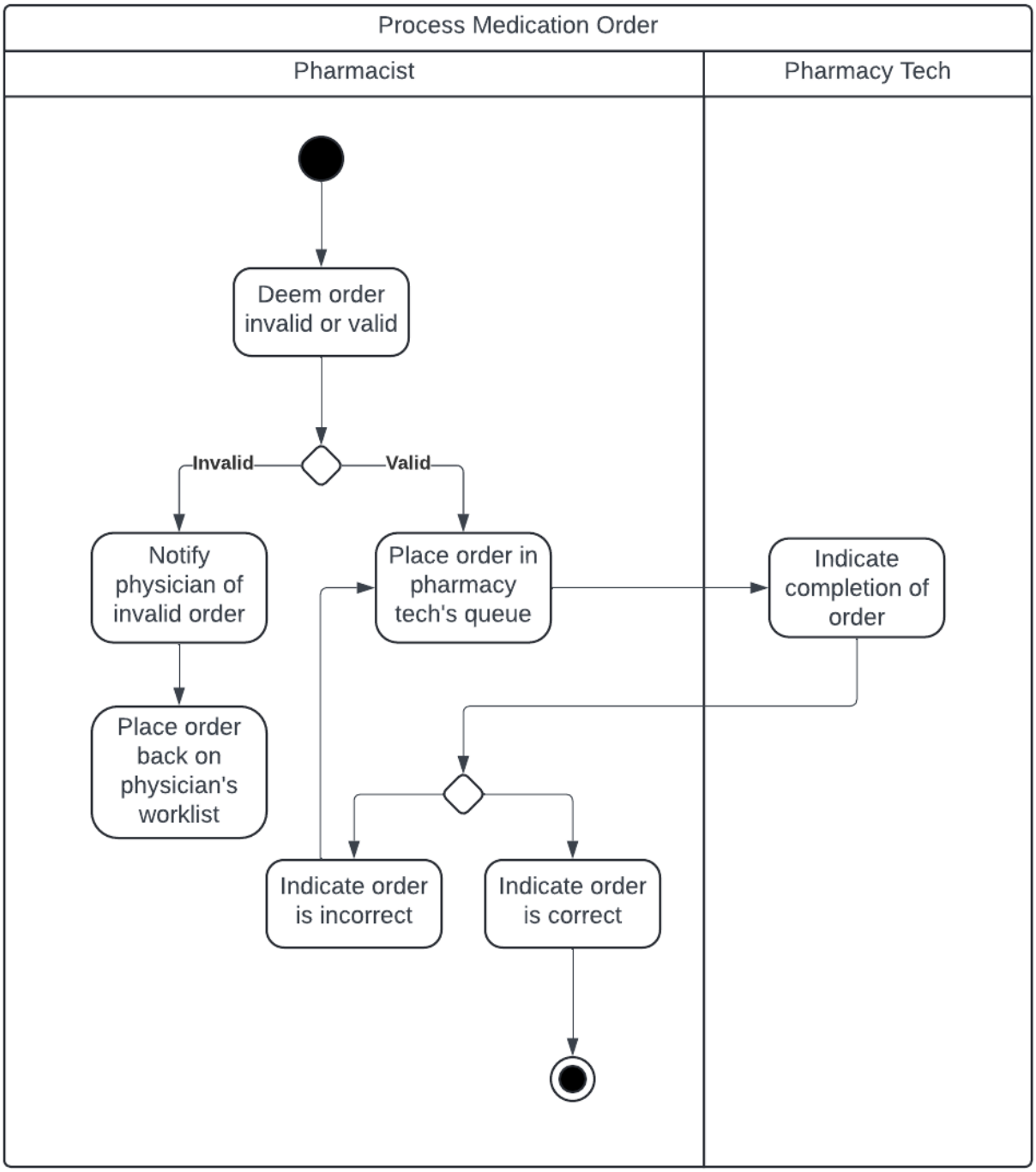

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Use Case Name: Process Medication Order	ID: 3	Importance Level: High
Primary Actor: Pharmacist	Use Case Type: Detail, Essential	
Stakeholders and Interests: Pharmacist – wants to improve the workflow and accuracy of processing medication orders		
Brief Description: This use case describes how a medication order is processed by the pharmacist and pharmacy technician.		
Trigger: The system sends a medication order to the pharmacy. Type: External		
Relationships: Association: Pharmacist and pharmacy technician Include: Extend: Generalization:		
Normal Flow of Events: 1. The pharmacist deems the medication order invalid or valid. If the order is valid, the system proceeds. If the order is invalid, S-1 subflow is performed. 2. The system places the order in the pharmacy technician's queue. 3. The tech indicates completion of the order. 4. The pharmacist indicates if the order is incorrect or correct. If the order is correct, the system proceeds. If the order is incorrect, the system returns to step #2.		
SubFlows: S-1: Invalid order 1. The system notifies the physician of an invalid order. 2. The system places the order back on the physician's worklist, and the system does not proceed to step #2 in the normal flow.		
Alternate/Exceptional Flows:		

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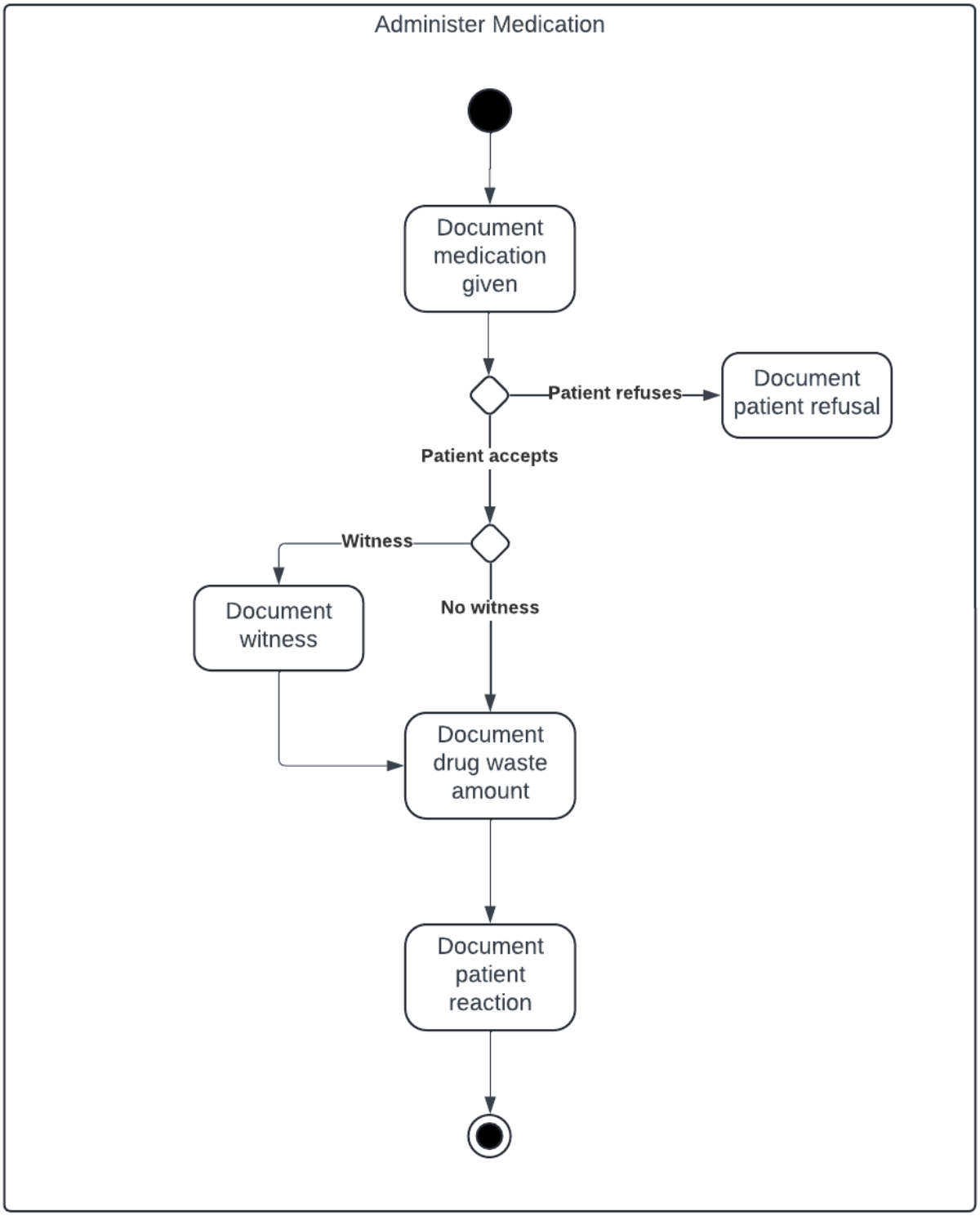

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Use Case Name: Administer Medication	ID: 4	Importance Level: High
Primary Actor: Nurse	Use Case Type: Detail, Essential	
Stakeholders and Interests: Nurse – needs to document details of medication administration due to being liable for the patient's care and lost medication Physician – needs to check patient compliance with prescribed medications		
Brief Description: This use case describes how medication administration is documented by the nurse.		
Trigger: The nurse administers the medication to the patient. Type: External		
Relationships: Association: Nurse Include: Extend: Generalization:		
Normal Flow of Events: 1. The nurse documents the medication given to the patient. 2. If the patient refuses the medication, the nurse documents patient refusal and does not continue to step #3. 3. If there is a witness present, the nurse documents the witness. 4. The nurse documents the amount of medication waste. 5. The nurse documents the patient interaction to the medication.		
SubFlows:		
Alternate/Exceptional Flows:		

Reference

Dennis, A., Wixom, B., & Tegarden, D. (2020). *Systems Analysis and Design: An Object-Oriented Approach with UML* (6th ed.). Wiley Global Education US. https://reader.yuzu.com/books/9781119561217

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